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The Journal of Physiology

Beat-to-beat QT interval variability as a tool to detect the underlying cellular mechanisms of arrhythmias

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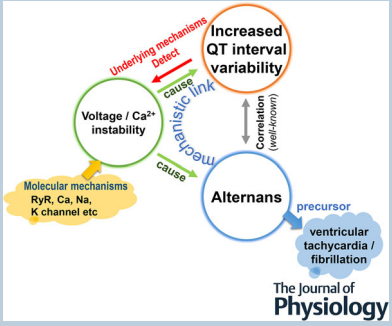
Handling Editors: Bjorn Knollmann & Axel Loewe

The peer review history is available in the Supporting Information section of this article (https://doi.org/10.1113/JP289051#support-information-section).



Abstract figure legend Mechanistic link between cardiac instability, QT interval variability (QTV), and alternans. Dynamical instability originating from either voltage or calcium handling drives both increased QTV and alternans (green arrows). Alternans is a well–established precursor to ventricular tachycardia and fibrillation. This mechanistic link can be used as a diagnostic tool. Measuring how QTV responds to pharmacological interventions that stabilize voltage or calcium dynamics may identify the predominant mechanism underlying arrhythmia risk (red arrow). QTV therefore represents a non-invasive tool for risk assessment and mechanism-specific therapy.

Abstract Increased beat-to-beat QT interval variability (QTV) on the electrocardiogram (ECG) has been associated with arrhythmia risk and sudden cardiac death. However, the underlying mechanisms driving increased QTV are not fully understood. Our previous work showed that membrane voltage instability is a major contributor to QTV. In this study, we investigated how intracellular calcium (Ca^{2+}) cycling instability is also a major contributor to QTV using a mathematical model of a ventricular myocyte that incorporates stochastic ion channel gating and detailed Ca^{2+} cycling. By independently modulating membrane voltage instability (via the L-type Ca^{2+} channel recovery time constant, τ_f) and intracellular Ca^{2+} cycling instability (via the steepness of the sarcoplasmic reticulum Ca^{2+} release-load relationship, u), we show that both voltage and Ca^{2+} instabilities significantly increase action potential duration (APD) variability, which contributes to QTV, even in the absence of overt arrhythmic patterns. Ca^{2+} transient variability increases with intracellular Ca^{2+} cycling instability, contributing to APD variability via Ca^{2+}-sensitive currents,

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DOI: 10.1113/JP289051

and consequently to QTV. Notably, APD variability/QTV significantly increases just before the onset of alternans, regardless of whether instability originates from voltage or Ca^{2+} dynamics. Thus, QTV may serve as a precursor to both voltage-driven and Ca^{2+}-driven alternans. Furthermore, pharmacological interventions that selectively stabilize voltage *vs.* Ca^{2+} cycling may selectively reduce QTV. These findings suggest that QTV can help distinguish between arrhythmias caused by electrical dysfunction and those caused by Ca^{2+} cycling dysfunction. Therefore, QTV has potential as a non-invasive tool not only to identify individuals at risk but also to predict the specific type and underlying cause of arrhythmias.

(Received 9 April 2025; accepted after revision 11 September 2025; first published online 4 October 2025)
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Key points

- Both membrane voltage and intracellular Ca^{2+} cycling instabilities contribute to increased QTV, even without overt arrhythmic patterns.
- Ca^{2+} transient variability increases with intracellular Ca^{2+} cycling instability and independently contributes to QTV, regardless of voltage instability.
- QTV serves as a precursor to both electrical and Ca^{2+} alternans, highlighting its potential as an early non-invasive marker for arrhythmic events.
- The response of QTV to specific pharmacological interventions may differentiate between voltage-driven and Ca^{2+}-driven instability, guiding personalized treatment strategies.
- The study suggests QTV as a promising tool for personalized arrhythmia risk assessment and mechanism-specific therapeutic strategies.

Introduction

Cardiac arrhythmia is often caused by dysfunction in intracellular calcium (Ca^{2+}) handling (Bers, 2014; Gyorke, 2009; Laurita & Rosenbaum, 2008; Lou et al., 2012; Pogwizd & Bers, 2002; ter Keurs et al., 2006). Under normal conditions, an action potential triggers Ca^{2+} influx via L-type Ca^{2+} channels (LTCCs) and subsequent Ca^{2+} release from the sarcoplasmic reticulum (SR) through ryanodine receptors (RyR) via the Ca^{2+}-induced Ca^{2+} release process, leading to muscle contraction (Bers, 2001; Bers, 2008; Chelu & Wehrens, 2007; Cheng et al., 1993; George, 2008). However, under pathological conditions, intracellular Ca^{2+} handling is often disrupted, which can lead to Ca^{2+} alternans, spontaneous Ca^{2+} release, waves, delayed afterdepolarizations (DADs) and triggered activity, all of which may contribute to the development of arrhythmias (January & Fozzard, 1988; Lipp & Pott, 1988; Schlotthauer & Bers, 2000; Tseng & Wit, 1987; Verkerk et al., 2001; Voigt et al., 2012). This understanding has been established over decades through various animal experiments and studies on human cells and tissues. However, it currently remains challenging to directly measure Ca^{2+} activity in patients. This raises a critical question: how can we predict clinically relevant Ca^{2+}-related arrhythmias before their onset? While an electrocardiogram (ECG) provides valuable information about electrical activity in the heart, it does not directly reveal information about Ca^{2+} activity. Can we retrieve information about Ca^{2+} activity from ECG recordings?

Increased beat-to-beat QT interval variability (QTV), a measure of beat-to-beat fluctuations in the QT interval on the ECG, is strongly associated with arrhythmias and sudden cardiac death (Atiga et al., 1998; Baumert et al., 2016; Floré et al., 2011; Hinterseer et al., 2008; Hinterseer et al., 2009; Hinterseer et al., 2010; Oosterhoff et al., 2011; Piccirillo et al., 2007; Smoczyńska et al., 2020; Tereshchenko & Berger, 2011; Thomsen et al., 2004, 2005, 2007). Clinically, an elevated magnitude of QTV, for instance, a standard deviation of QT intervals (SDQT) greater than approximately 5 ms, has been identified as an indicator of increased arrhythmic risk (Baumert et al., 2016). However, the underlying mechanisms driving increased QTV remain poorly understood.

Ion channel gating stochasticity can be a source of electrical variability (Baumert et al., 2016; Heijman et al., 2013). However, since there are thousands of channels (e.g. 50,000∼500,000 sodium (Na^+) channels, 10,000∼100,000 LTCCs and ∼2 million RyR channels) at the single-cell level, the variability at the cellular level is much smaller than that of individual ion channels according to the law of large numbers. In addition, since cells are electrotonically

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coupled via gap junctions in tissue, this variability is further reduced at the tissue scale (Heijman et al., 2013). This indicates that ion channel stochasticity alone cannot fully explain the observed magnitude of QTV on the ECG. The persistence of significant QTV at the ECG level suggests that factors beyond ion channel gating stochasticity must be involved.

We have recently shown that dynamical instability is a major contributor to action potential duration (APD) variability (Sato, Hegyi et al., 2025). Our research demonstrates a mechanistic link between APD variability and the slope of the APD restitution curve, which indicates the instability of the cardiac membrane potential. In general, as instability increases, APD variability increases even during normal periodic action potentials. This cellular APD variability, in turn, contributes to the QTV observed on the ECG. This effect becomes particularly pronounced near the onset of alternans, where APD variability increases substantially. These results suggest that increased APD variability, and consequently QTV, can serve as an indicator of arrhythmic risk even in healthy individuals without overt arrhythmic ECG patterns.

Intracellular Ca^{2+} dynamics can be unstable, independent of membrane voltage, and induce irregular voltage activity through Ca^{2+}-sensitive currents such as the Na^{+}–Ca^{2+} exchanger (NCX) and I_{CaL}. An example is that cardiac alternans (in both APD and contraction strength) can be initiated by the refractory state of the RyR, and the instigating Ca^{2+} transient alternans can occur even when the voltage waveform is clamped (Picht et al., 2006; Wang et al., 2014, 2021). However, the influence of intracellular Ca^{2+} cycling instability on APD variability, and thus its contribution to QTV, remains unknown.

In this study, we will show that intracellular Ca^{2+} transients exhibit increased variability as the intracellular Ca^{2+} cycling becomes unstable, contributing to APD variability, and thus QTV, independent of voltage dynamics. Furthermore, we propose that QTV sensitivity to specific pharmacological interventions can provide insight into the underlying mechanism of arrhythmia risk.

Methods

Ethical statement

This is a computational study that did not involve the use of animals or human samples, and therefore, ethical approval was not required.

Mathematical model

In this study, we used a mathematical model of the ventricular myocyte action potential and intracellular Ca^{2+} cycling developed by Shiferaw et al. (2005). We chose this model because it allows for the independent modulation of membrane voltage instability and intracellular Ca^{2+} cycling instability with limited interference between them. This model incorporates detailed voltage dynamics with ionic currents (I_{Na}, I_{CaL}, I_{Kr}, I_{Ks}, I_{to}, I_{K1}, etc.), detailed intracellular Ca^{2+} handling (SR Ca^{2+} uptake, release and buffering). The membrane voltage is governed by

$$\frac{dV_m}{dt} = -\frac{1}{C_m}(I_{Na} + I_{to} + I_{Kr} + I_{Ks} + I_{Kp} + I_{K1} + I_{NaCa} + I_{CaL} + I_{stim}),$$

where C_m is the cell membrane capacitance, I_{Na} is the fast sodium current, I_{to} is the transient outward potassium current, I_{Kr} and I_{Ks} are the rapid and slow components of the delayed rectifier potassium current, I_{Kp} is the plateau potassium current, I_{K1} is the inward rectifier potassium current, I_{NaCa} is the NCX, and I_{CaL} is the L-type Ca^{2+} current, and I_{stim} is the stimulus current.

Stochastic ion channel gating

Stochasticity was introduced into the model by incorporating noise into the LTCC gating, which was simulated using the Langevin equation (Fox, 1997; Fox & Lu, 1994). The LTCC flux (J_{CaL}) is given by

$$J_{CaL} = -g_{Ca} \cdot d \cdot f \cdot f_{Ca} \cdot i_{Ca},$$

where g_{Ca} is the maximum conductance of J_{CaL}, d is the activation gating variable, f is the voltage-dependent inactivation gating variable and f_{Ca} is the Ca^{2+} dependent inactivation gating variable, and i_{Ca} is the single-channel current.

The single channel current (i_{Ca}) is given by

$$i_{Ca} = \frac{4V_m F^2}{RT}\left(\frac{c_s e^{2a} - 0.34[Ca^{2+}]_O}{e^{2a} - 1}\right)$$

with $a = V_m\ F/RT$ and c_s is the Ca^{2+} concentration in the submembrane space.

Stochasticity was applied to the d-gate, the f-gate, and the f_{Ca}-gate of the LTCC. The deterministic version of the gating equations are:

$$\frac{dd}{dt} = \alpha_d\ (1 - d) - \beta_d d,$$

$$\frac{df}{dt} = \alpha_f\ (1 - f) - \beta_f f,$$

$$\frac{df_{Ca}}{dt} = \alpha_{f_{Ca}}\ (1 - f_{Ca}) - \beta_{f_{Ca}} f_{Ca},$$

where d, f and f_{Ca} are the gating variables, α_d, α_f, $\alpha_{f_{Ca}}$ are the opening rates and β_d, β_f, $\beta_{f_{Ca}}$ are the closing rates.

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The corresponding stochastic versions of these equations are:

$$\frac{\mathrm{d}d}{\mathrm{d}t} = \alpha_d \ (1-d) - \beta_d d + \eta_d(t),$$

$$\frac{\mathrm{d}f}{\mathrm{d}t} = \alpha_f \ (1-f) - \beta_f f + \eta_f(t),$$

$$\frac{\mathrm{d}f_{\mathrm{Ca}}}{\mathrm{d}t} = \alpha_{f_{\mathrm{Ca}}} \ (1-f_{\mathrm{Ca}}) - \beta_{f_{\mathrm{Ca}}} f_{\mathrm{Ca}} + \eta_{f_{\mathrm{Ca}}}(t),$$

where $\eta_d(t)$, $\eta_f(t)$, $\eta_{f_{\mathrm{Ca}}}(t)$ are the noise terms that satisfy the following correlation functions:

$$\langle \eta_d(t)\,\eta_d(t')\rangle = \frac{\alpha_d(1-d)+\beta_d d}{N}\,\delta(t-t'),$$

$$\langle \eta_f(t)\,\eta_f(t')\rangle = \frac{\alpha_f(1-f)+\beta_f f}{N}\,\delta(t-t'),$$

$$\langle \eta_{f_{Ca}}(t)\,\eta_{f_{Ca}}(t')\rangle = \frac{\alpha_{f_{Ca}}(1-f_{Ca})+\beta_{f_{Ca}} f_{Ca}}{N}\delta(t-t'),$$

where N is the number of LTCCs. In this study, we chose $N = 100{,}000$, which is a physiologically realistic number in an adult ventricular myocyte. We note that our main results are qualitatively the same regardless of the choice of ion channel (Sato, Hegyi et al., 2025).

Simulation protocol

Cells were paced at a cycle length (PCL) of 300 ms. Simulations were run until a steady state was achieved, typically requiring 300∼1000 beats, with more beats near bifurcation points. This PCL (300 ms) is relatively fast for the rabbit ventricular myocyte. Experimentally, alternans has been observed at this rate or even slower rates under various conditions, including the effects of drugs. Past studies also demonstrate electrical and Ca^{2+} alternans in isolated myocytes and tissue at similar PCLs, particularly in pathological conditions such as Ca^{2+} overload or reduced repolarization reserve (Picht et al., 2006; Wang et al., 2014, 2021). Furthermore, the alternans observed in our simulations under unstable conditions (induced by varying τ_f or u as described in detail below) are consistent with alternans reported in diseased myocytes and patients (Narayan et al., 2007).

There are multiple ways to quantify QTV (Baumert et al., 2016). In this study, APD and Ca^{2+} transient variability were quantified as the standard deviation of APD and Ca^{2+} transient peaks, respectively. This variability was calculated over 10,000 consecutive beats after the cell reached a steady state. With 10,000 beats, a typical standard error of the mean (SEM) is 0.01∼0.1 ms.

2D tissue simulations

We also simulated a monodomain, two-dimensional (2D) sheet consisting of 200 × 200 cells (3 cm × 3 cm) with a spatial discretization of 150 μm. Cells were electrotonically coupled via gap junctions, and the membrane voltage in each cell was governed by the reaction–diffusion equation. Gap junctional coupling was represented by a diffusion coefficient, $D = 0.001$ cm^2/ms, producing an isotropic conduction velocity of approximately 85 cm/s, which is physiologically realistic for ventricular tissue (Kléber & Rudy, 2004). No-flux boundary conditions were applied at all tissue edges.

A pseudo-ECG was calculated from the 2D tissue simulations to approximate the surface ECG. Variability of the QT-like interval on the pseudo-ECG was then analysed.

Numerical methods

All simulation codes were written in C/C++ and are available on GitHub (https://doi.org/10.5281/zenodo.16790275). The simulations were performed on a workstation equipped with an Intel Core i9-13900K CPU and 128 GB RAM, running Ubuntu 24.04. The ordinary differential equations were solved using the forward Euler method with a time step of 0.05 ms. For 2D tissue simulations, the reaction–diffusion equation was solved using the operator splitting method (Qu & Garfinkel, 1999).

Results

Cardiac cells can be unstable due to instability of the membrane voltage, instability of intracellular Ca^{2+} cycling, or both (Sato et al., 2007; Shiferaw et al., 2005; Shiferaw et al., 2006; Weiss et al., 2006). In this study, we varied the recovery time constant of the LTCC (τ_f) to modulate membrane voltage instability. We selected τ_f because it primarily affects voltage stability while minimally influencing Ca^{2+} instability (Sato, Hatano et al., 2025). (For example, altering LTCC conductance also changes voltage instability, but it simultaneously affects Ca^{2+} instability.) On the other hand, to modulate Ca^{2+} instability, we varied the steepness of the SR Ca^{2+} release-*versus*-load relationship (u) (release gain). This parameter predominantly affects Ca^{2+} instability with limited effects on membrane voltage instability. Figure 1 shows the stability boundary when these parameters (τ_f, u) are varied. Representative AP and Ca^{2+} transient traces for the stable condition, as well as for conditions of increased τ_f (voltage instability) and increased u (Ca^{2+} instability), are shown in Appendix Fig. A1*A*–*C*.

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APD variability increases as membrane voltage instability increases

We first measured APD variability as a function of the recovery time constant of the LTCC (τ_f) (Fig. 2*A*, left axis). As τ_f increases, the APD restitution curve becomes steeper. The slope of the APD restitution curve is indicated on the right axis. Both APD variability and restitution curve slope increase as τ_f increases. Therefore, APD variability (which contributes to QTV) and the slope of the APD restitution curve, which indicates membrane voltage instability, are positively correlated. The blue arrow in the figure indicates the onset of alternans. Notably, APD variability increases even when action potentials (APs) are periodic (showing no alternans or other arrhythmic patterns). Figure 2*B* shows APD *versus* beat number at different τ_f values indicated in Fig. 2*A*. These results confirm that membrane voltage instability, as reflected in the slope for the APD restitution curve, is a major contributor to APD variability (Sato, Hegyi et al., 2025).

Ca^{2+} transient variability increases as Ca^{2+} cycling becomes unstable

Intracellular Ca^{2+} cycling has its own dynamics and can become unstable independently of voltage instability (Chudin et al., 1999; Picht et al., 2006; Wang et al., 2014, 2021). We paced the cell using an AP waveform clamp, which eliminates APD variability and thus Ca^{2+} transient variability arises solely from intracellular Ca^{2+} cycling. We then increased the steepness of the SR Ca^{2+} release *versus* load relationship (u). Ca^{2+} transient variability increases with the steepness of this relationship (Fig. 3*A*).

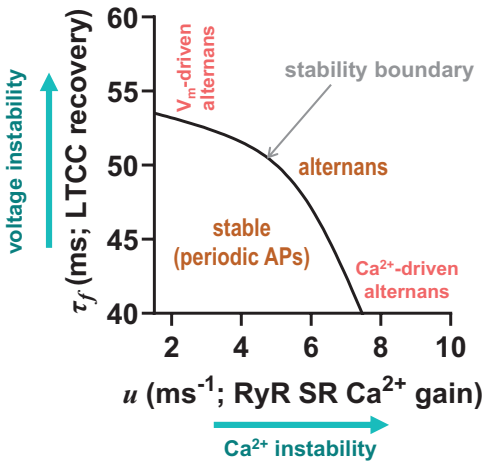

Figure 1. Stability boundary
Alternans can be caused by membrane voltage instability, Ca^{2+} cycling instability, or both. We varied the recovery time constant of the LTCC (τ_f) to modulate membrane voltage instability. To modulate Ca^{2+} instability, we varied the steepness of the SR Ca^{2+} release-*versus*-load relationship (u).

These results suggest that Ca^{2+} transient variability and intracellular Ca^{2+} instability are positively correlated. Moreover, this result demonstrated that Ca^{2+} transient variability can increase independently of membrane voltage variability. The arrow indicates the onset of alternans. Figure 3*B* shows Ca^{2+} peak *versus* beat number at different u values.

Ca^{2+} transient variability contributes to APD variability independently of membrane voltage instability

The same cell as shown in Fig. 3*A* was paced using current-clamp mode. Since τ_f is small, the membrane voltage dynamics remain stable. We varied the steepness of the SR Ca^{2+} release *versus* load relationship (u). APD variability increases with increasing steepness of this relationship (Fig. 3*C*). This result suggests that Ca^{2+} transient variability, which comes from intracellular Ca^{2+} instability as demonstrated in Fig. 3*A* and *B*, plays a critical role in APD variability and consequently contributes to QTV. Figure 3*D* shows APD *versus* beat number at different u values. This demonstrates that Ca^{2+} cycling instability can directly contribute to APD variability,

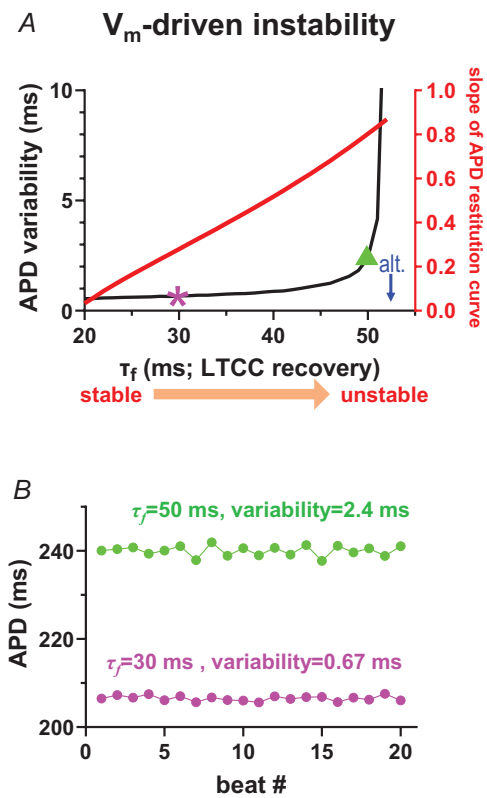

Figure 2. APD variability and instability (voltage-driven case)
A, left axis: APD variability *vs.* τ_f. u = 3.0 ms^{-1}. Right axis: slope of the APD restitution curve *vs.* τ_f. *B*, sample traces for τ_f = 30 and 50 ms. APD *vs.* beat number.

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independent of membrane voltage instability and happens at lower u level without AP-clamp.

Variability reaches its maximum at the onset of alternans when every other beat is used

In our previous study, we have shown that APD variability reaches its maximum at the onset of alternans when calculated using every other beat (Sato, Hegyi et al., 2025). That study focused solely on voltage instability. In the present study, the cell has both membrane voltage instability and Ca^{2+} instability.

First, we calculated APD variability using every other beat for the voltage-driven instability. APD variability reaches its maximum at the onset of alternans when τ_f and thus the voltage instability was varied (Fig. 4*A*).

Ca^{2+} instability also can cause alternans independently of voltage instability (Chudin et al., 1999; Picht et al., 2006; Sato, Hatano et al., 2025; Wang et al., 2014, 2021). We varied the steepness of the SR Ca^{2+} release *versus* load relationship (u) and paced using an action potential waveform clamp. In this case, action potentials are periodic with no alternans. However, Ca^{2+} transients can exhibit alternans as u and thus Ca^{2+} instability becomes larger (Fig. 4*B*, top). We also measured Ca^{2+} transient variability using every other beat. Ca^{2+} variability reaches its maximum at the onset of Ca^{2+} alternans (Fig. 4*B*, bottom).

We then paced the cell using the current clamp mode and measured the bifurcation diagram (Fig. 4*C*, top) and APD variability using every other beat (Fig. 4*C*, bottom). APD variability reaches its maximum at the onset of alternans (and again occurs at lower u when APD can also vary, indicative of synergy).

These results suggest that APD variability and thus QTV reaches its maximum at the onset of alternans when calculated using every other beat, regardless of the underlying instability mechanisms.

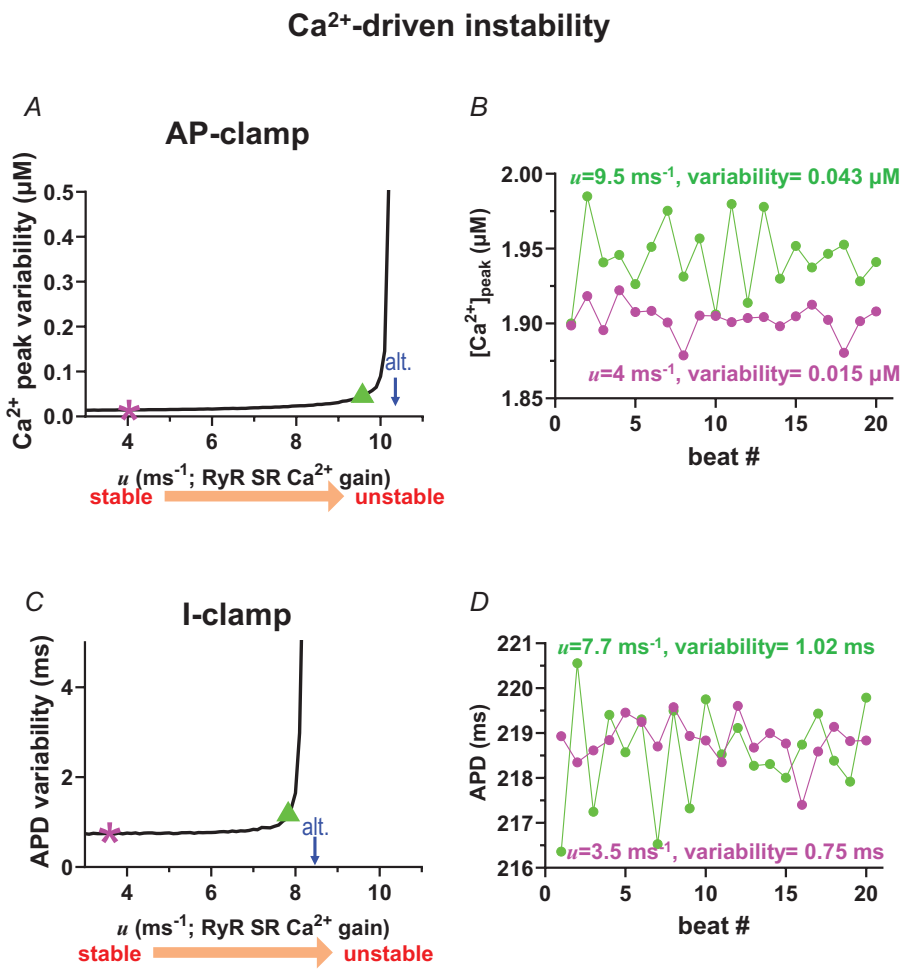

Figure 3. APD variability and instability (Ca^{2+}-driven case)
A, Ca^{2+} peak *vs.* u (action potential waveform clamp). *B*, sample traces for u = 4 and 9.5 ms^{-1}. Ca^{2+} peak *vs.* beat number. *C*, APD variability *vs.* u (current clamp). *D*, sample traces for u = 3.5 and 7.7 ms^{-1}. APD *vs.* beat number.

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Tissue-level variability

To understand how single-cell APD variability translates to tissue-level QTV, we performed 2D tissue simulations and calculated pseudo-ECGs (Appendix Fig. A2*A* and *B*). These simulations demonstrate that variability observed on the pseudo-ECG, reflecting beat-to-beat fluctuations in the QT interval, increases as parameters (e.g. τ_f or u) approach bifurcation points. While intercellular coupling in the tissue acts to reduce the absolute magnitude of APD variability compared to isolated single cells as predicted by Heijman et al. (2013), the tissue-level variability (reflected in the pseudo-ECG) still significantly increases near these

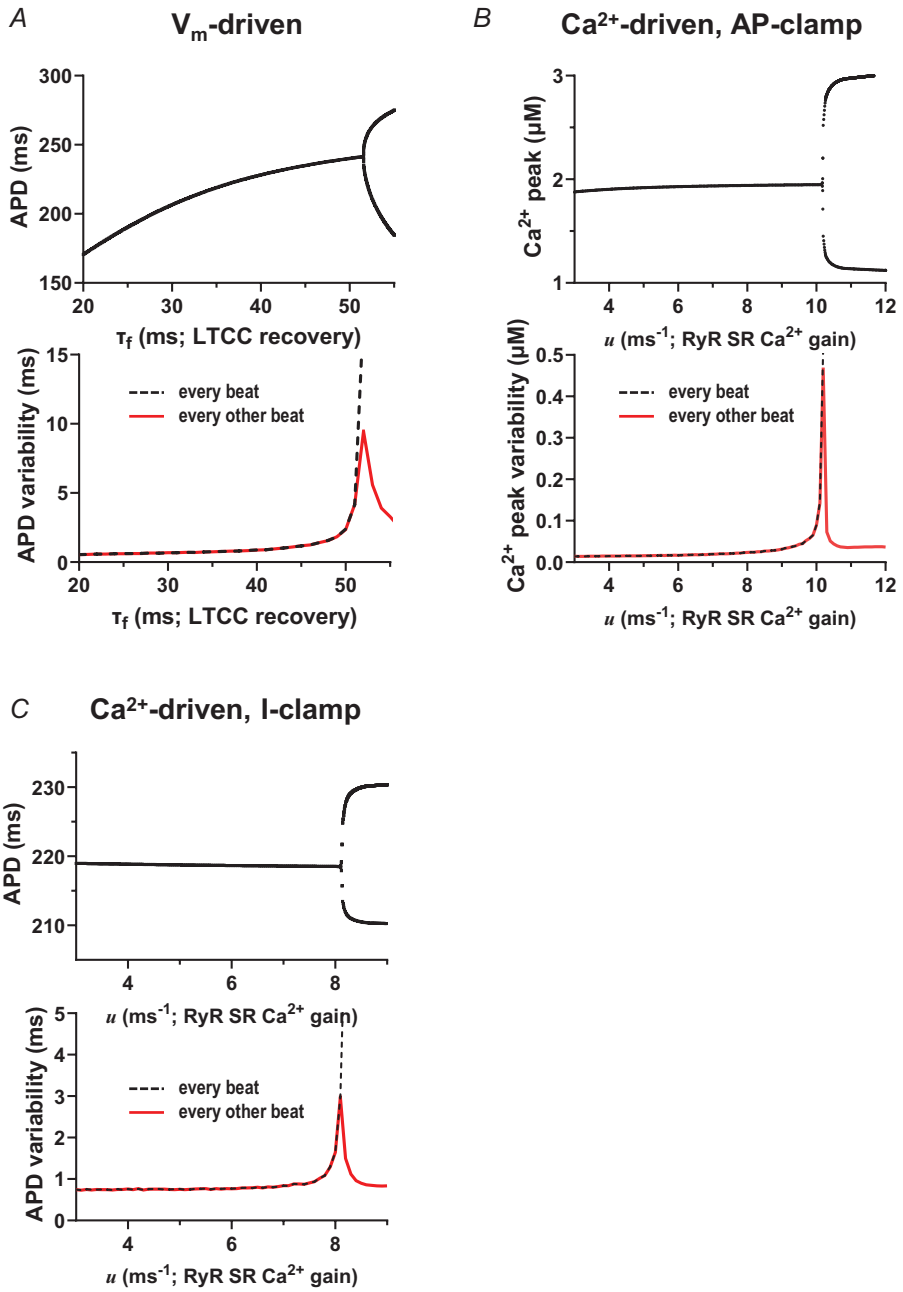

Figure 4. APD variability using every other beat and the onset of alternans
A, top: bifurcation diagram for voltage-driven instability. APD *vs.* τ_f. u = 3.0 ms^{-1}. Bottom: corresponding APD variability using every beat (black) and APD variability using every other beat (red). *B*, top: bifurcation diagram for Ca^{2+}-driven instability (action potential waveform clamp). Ca^{2+} peak *vs.* u. τ_f = 35 ms. Bottom: corresponding APD variability using every beat (black) and APD variability using every other beat (red). *C*, top: bifurcation diagram for Ca^{2+}-driven instability (current clamp). APD *vs.* u. τ_f = 35 ms. Bottom: corresponding APD variability using every beat (black) and APD variability using every other beat (red).

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bifurcation points (Appendix Fig. A2*C* and *D*). This occurs because inherent fluctuations persist longer and are amplified as parameters move closer to a bifurcation (Sato, Hegyi et al., 2025). This principle holds true even in coupled tissue. These findings from multicellular tissue simulations support the hypothesis that QTV on ECG can reflect underlying cellular instabilities.

QTV as a tool to identify the underlying mechanisms

APD variability, and consequently its contribution to QTV, increases as instability increases even when action potentials are periodic (Figs 2*A* and 3*C*). In this study, we investigated how APD variability responds to parameter modifications that conceptually mimic the effects of pharmacological interventions. This was done by varying the maximal ion channel conductances and measuring the corresponding change in APD variability. Specifically, to alter the stability of intracellular Ca^{2+} cycling, we varied the RyR conductance (G_{RyR}), as would occur with agents such as dantrolene, flecainide or caffeine. Conversely, to alter the stability of the membrane voltage system, we varied the conductance of key repolarizing potassium currents (G_{Kr} and G_{Ks}), analogous to the action of K^+ channel blockers, or potential enhancers (or discontinuation of K^+ channel inhibitors). These targeted modifications allowed us to assess how selectively addressing one source of instability over the other influences the overall APD variability. When the RyR conductance (G_{RyR}) was reduced by 50% (simulating RyR stabilization), APD variability decreased, especially in Ca^{2+}-driven APD variability cells (Fig. 5*A*). Ca^{2+}-driven APD variability is defined as APD variability arising primarily from intracellular Ca^{2+} cycling instability (steep SR Ca^{2+} release *vs*. load relationship, small τ_f, large u). As shown in Figs 2–4, APD variability significantly increases near the onset of alternans. Reducing G_{RyR} stabilizes intracellular Ca^{2+} cycling, shifting the onset of alternans toward a more stable state. Consequently, the cellular state moves further from the alternans onset, leading to a reduction in APD variability (Fig. 6*B* and *D*). In contrast, reducing G_{RyR} has minimal impact on membrane voltage stability and only slightly shifts the onset of alternans. As a result, APD variability remains largely unchanged (Fig. 6*A* and *C*).

Conversely, when G_{Kr} (Figs 5*B* left and 7) or G_{Ks} (Figs 5*B* right and 8) was increased by 200%, APD variability decreased, particularly in voltage-driven APD variability cells (although APD variability was also reduced in Ca^{2+}-driven cells). Voltage-driven APD variability is defined as APD variability arising primarily from membrane voltage instability (steep APD restitution mechanism, large τ_f, small u).

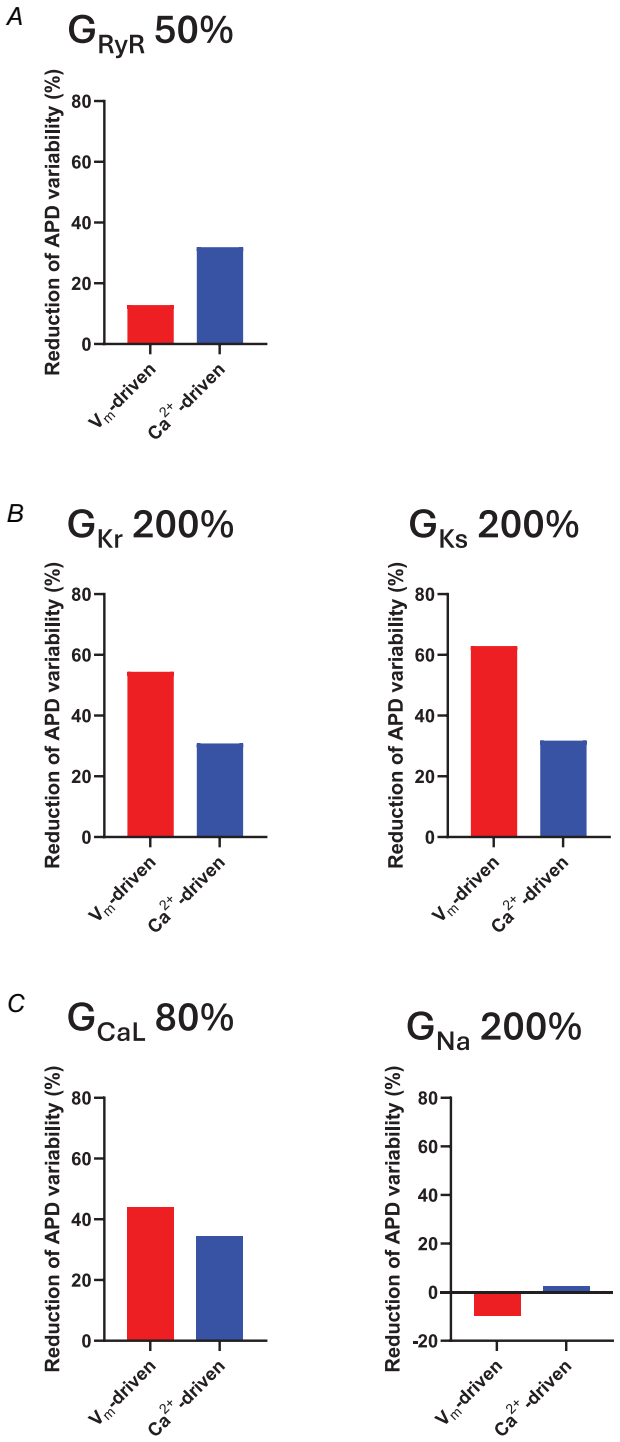

Figure 5. Reduction in APD variability when ion channel conductances are varied
A, G_{RyR} 50%. APD variability was significantly reduced in Ca^{2+}-driven instability cell. *B*, left: G_{Kr} 200%. APD variability was significantly reduced in voltage-driven instability cell. Right: G_{Ks} 200%. APD variability was significantly reduced in voltage-driven instability cell. *C*, left: G_{CaL} 80%. APD variability was reduced in both cells. In this case, we cannot distinguish the underlying instability mechanism. Right: G_{Na} 200%. APD variability was almost unchanged in both cases. We also cannot use this to distinguish underlying instability mechanism.

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Reducing the SR Ca^{2+} release current stabilizes Ca^{2+} cycling while reducing I_{Kr} or I_{Ks} flattens the APD restitution curve. The response depends on whether the intervention stabilizes Ca^{2+} cycling or membrane voltage. If G_{CaL} is reduced, both Ca^{2+} cycling and membrane voltage will be stabilized, leading to a reduction in APD variability in both cases (Figs 5*C* left and 9). On the other hand, if G_{Na} is increased, it has minimal effects on both voltage and Ca^{2+} instability, resulting in only a limited change in APD variability in both cases (Figs 5*C* right and 10).

Discussion

In this study, we investigated the relationship between QTV and the underlying cellular mechanisms that contribute to it, primarily by examining APD variability.

The association between increased QTV and arrhythmia risk has been established in numerous clinical studies (Atiga et al., 1998; Baumert et al., 2016; Floré et al., 2011; Hinterseer et al., 2008, 2009, 2010; Oosterhoff et al., 2011; Piccirillo et al., 2007; Smoczyńska et al., 2020; Tereshchenko & Berger, 2011; Thomsen et al., 2004, 2005, 2007). However, the cellular mechanisms underlying QTV remain incompletely understood. Our recent work demonstrates that while stochastic ion channel gating contributes to electrical variability, the magnitude of this variability is insufficient to account for the clinically observed QTV due to the law of large numbers and electrotonic coupling between cells. Instead, our results pointed to dynamical instability as a major contributor to APD variability (Sato, Hegyi et al., 2025), which in turn can manifest as QTV.

In clinical settings, only ECG (voltage activity) can be measured. There are no direct methods to assess Ca^{2+} activity non-invasively in the human heart. In this study, we showed that fluctuations in Ca^{2+} transients significantly contribute to APD variability independently of membrane voltage instability. Our findings provide novel insights into the cellular mechanisms underlying QTV and suggest potential applications for using QTV as a non-invasive marker of arrhythmia risk. Furthermore, our results demonstrate that QTV may serve as a valuable tool for detecting the predominant underlying cellular mechanisms of arrhythmias.

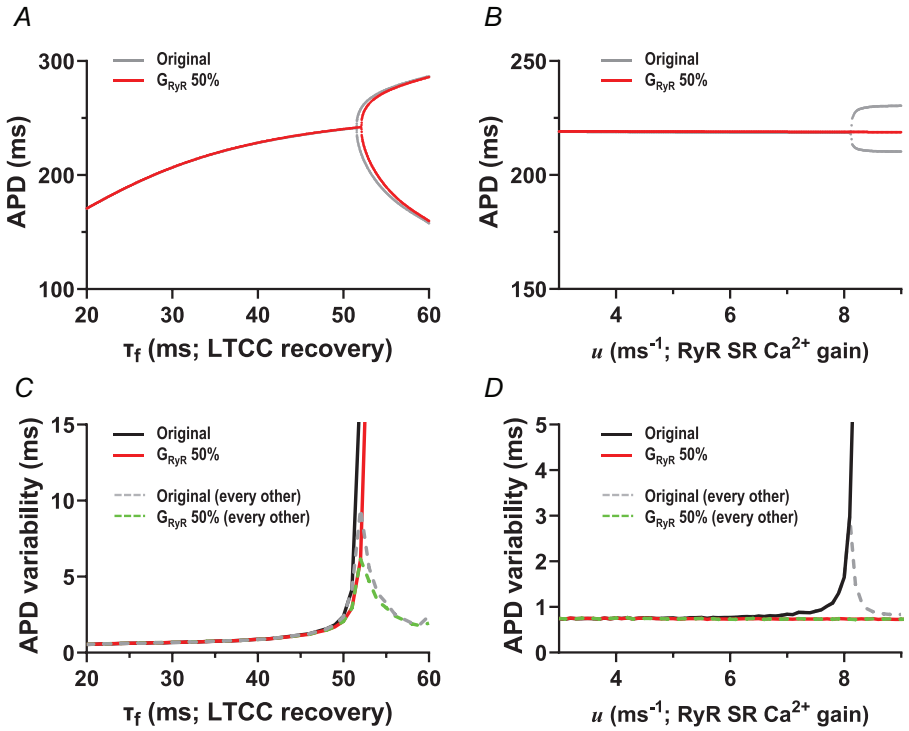

Figure 6. Bifurcation diagram and APD variability for G_{RyR} 50%
A, bifurcation diagram for voltage-driven instability. APD *vs.* τ_f. $u = 3.0\ ms^{-1}$. Grey: original value, red: G_{RyR} 50%. *B*, bifurcation diagram for Ca^{2+}-driven instability. APD *vs.* u. $\tau_f = 35$ ms. *C*, APD variability *vs.* τ_f for voltage-driven instability. Black: original value; red: G_{RyR} 50%; grey: APD variability using every other beat, original value; green: APD variability using every other beat, G_{RyR} 50%. *D*, APD variability *vs.* τ_f for Ca^{2+}-driven instability. Black: original value; red: G_{RyR} 50%, Grey: APD variability using every other beat, original value; green: APD variability using every other beat, G_{RyR} 50%.

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APD variability/QTV as a precursor to alternans

Alternans is a precursor to cardiac arrhythmias and is strongly linked to the initiation of lethal arrhythmias and sudden cardiac death (Fox et al., 2002; Garfinkel et al., 2000; Groenendaal et al., 2014; Hayashi et al., 2007; Pastore et al., 1999). Our study demonstrates that increased APD variability/QTV can serve as an early predictor of both electrical and Ca^{2+} alternans – essentially acting as a 'precursor to the precursor (i.e. alternans)'.

Our results show that APD variability can be elevated even when action potentials appear periodic (i.e. without change in heart rate variability), suggesting that QTV serves as a marker of arrhythmic risk before the onset of alternans. This underscores the potential of QTV analysis to predict alternans before it becomes apparent on ECG.

Our previous work established that APD variability is not solely driven by stochastic ion channel gating but is primarily due to instability (Sato, Hegyi et al., 2025). That study focused on voltage instability alone. In the present study, we extend this understanding by demonstrating that intracellular Ca^{2+} dynamics significantly contribute to APD variability. Our findings indicate that the APD variability that underlies QTV can arise from both membrane voltage instability and intracellular Ca^{2+} cycling instability.

Using our modelling approach, we dissected the relative contributions of these two mechanisms and showed that Ca^{2+} cycling instability can directly influence APD variability, independent of voltage dynamics. Specifically, we found that variability in Ca^{2+} transients increases with the steepness of the SR Ca^{2+} release *vs.* load relationship, which in turn significantly amplifies APD variability via Ca^{2+}-sensitive currents.

In this study, APD variability increases with τ_f, which correlates with the slope of the APD restitution curve (Fig. 2). While our prior study varied pacing cycle length (PCL) to induce instability, in this study, we varied τ_f to explore its effects. This confirms our previous finding that instability is a major driver of APD variability, regardless of parameter choice (τ_f, PCL, etc.).

Ca^{2+} transient variability also increases with the steepness of the SR Ca^{2+} release *vs.* load relationship

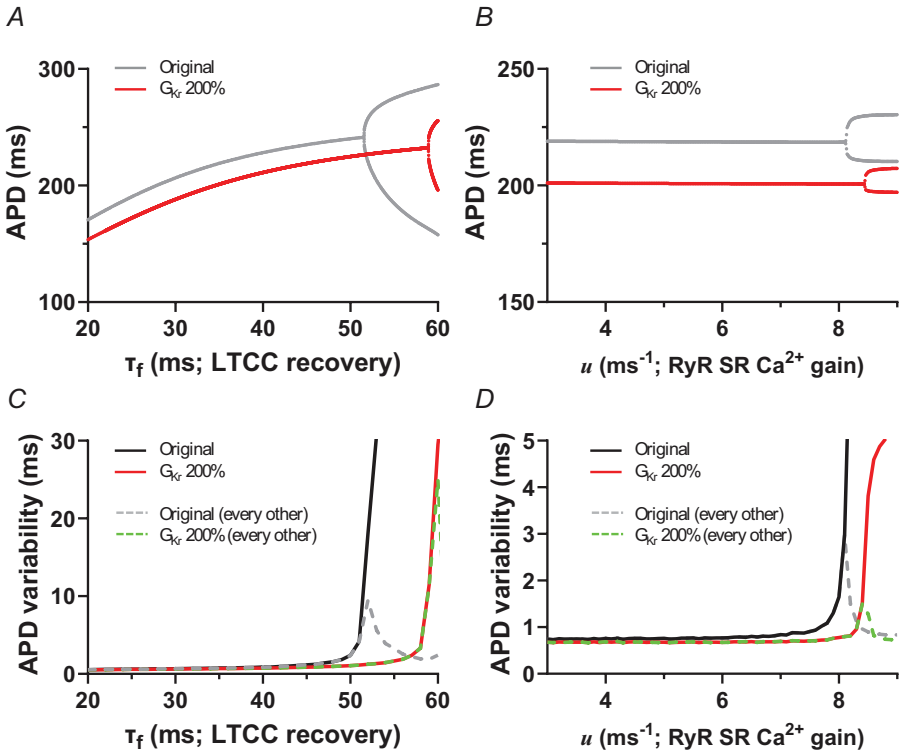

Figure 7. Bifurcation diagram and APD variability for G_{Kr} 200%
A, bifurcation diagram for voltage-driven instability. APD *vs.* τ_f. $u = 3.0\ ms^{-1}$. Grey: original value; red: G_{Kr} 200%. *B*, bifurcation diagram for Ca^{2+}-driven instability. APD *vs.* u. $\tau_f = 35$ ms. *C*, APD variability *vs.* τ_f for voltage-driven instability. Black: original value; red: G_{Kr} 200%; grey: APD variability using every other beat, original value; green: APD variability using every other beat, G_{Kr} 200%. *D*, APD variability *vs.* τ_f for Ca^{2+}-driven instability. Black: original value; red: G_{Kr} 200%; grey: APD variability using every other beat, original value; green: APD variability using every other beat, G_{Kr} 200%.

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(Fig. 3). Notably, this Ca^{2+}-driven variability persists even when membrane voltage dynamics are stabilized, such as under action potential waveform clamp conditions (Fig. 3*B*). This highlights that intracellular Ca^{2+} cycling can be an independent source of APD variability. In other words, Ca^{2+} instability alone can cause large APD variability via Ca^{2+}-sensitive currents, independent of intrinsic voltage instability (Fig. 3*C*).

In both cases, the increase in variability is most pronounced near the onset of alternans (Figs 2 and 3). This aligns with the mathematical theories of bifurcation with stochastic noise (Kuehn, 2011; Quail et al., 2015).

It is empirically known that increased QTV precedes arrhythmia. By linking microscopic dysregulation at the molecular/cellular scale to macroscopic ECG variability, our results bridge a critical gap in understanding QTV. We show that APD variability increases as both membrane voltage and intracellular Ca^{2+} cycling become unstable, even when action potentials appear periodic. This suggests that QTV can serve as an early indicator of arrhythmic risk, potentially before overt arrhythmic patterns manifest on ECG.

Onset of alternans and variability using every other beat

In our previous study, we demonstrated that APD variability reaches its maximum when calculated using every other beat (Sato, Hegyi et al., 2025). In this study, we extend this finding by showing that APD variability, and consequently QTV, also reaches its maximum when calculated using every other beat, regardless of whether the underlying mechanism is voltage-driven or Ca^{2+}-driven (Fig. 4). Furthermore, when the cell is paced using an action potential waveform clamp, Ca^{2+} variability similarly reaches its peak when calculated using every other beat (Fig. 4*B*).

Importantly, before the onset of alternans, APD variability calculated using every other beat yields values identical to those calculated using every beat (Fig. 4). This indicates that calculating QTV using every other beat does not interfere with current applications of QTV that rely on QTV using every beat. Moreover, our findings suggest that the onset of alternans can be identified by the peak of QTV calculated using every other beat, providing a novel and practical approach for detecting alternans development.

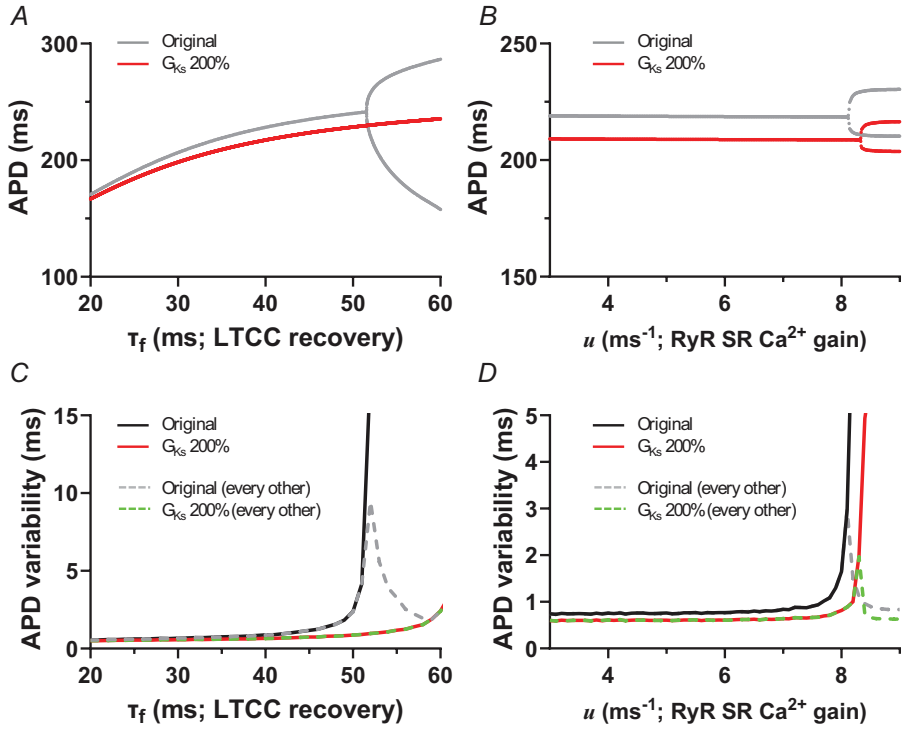

Figure 8. Bifurcation diagram and APD variability for G_{Ks} 200%
A, bifurcation diagram for voltage-driven instability. APD *vs.* τ_f. $u = 3.0$ ms^{-1}. Grey: original value; red: G_{Ks} 200%. *B*, bifurcation diagram for Ca^{2+}-driven instability. APD *vs.* u. $\tau_f = 35$ ms. *C*, APD variability *vs.* τ_f for voltage-driven instability. Black: original value; red: G_{Ks} 200%; grey: APD variability using every other beat, original value; green: APD variability using every other beat, G_{Ks} 200%. *D*, APD variability *vs.* τ_f for Ca^{2+}-driven instability. Black: original value; red: G_{Ks} 200%; grey: APD variability using every other beat, original value; green: APD variability using every other beat, G_{Ks} 200%.

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QTV as a tool to detect the underlying cellular mechanisms of arrhythmias

Another key translational implication of this study is the potential to use the pharmacological sensitivity of APD variability, and by extension QTV, to differentiate between predominantly voltage- and Ca^{2+}-driven instability in clinical settings, thereby enabling personalized risk stratification and treatment strategies. Our simulations demonstrate that interventions targeting specific mechanisms differentially modulate APD variability depending on the underlying instability (Figs 5–10).

Specifically, reducing RyR conductance, which stabilizes Ca^{2+} cycling by reducing SR Ca^{2+} release, significantly reduced APD variability in Ca^{2+}-driven instability cells (induced by increased steepness of the SR Ca^{2+} release *vs.* load relationship, u). Conversely, increasing I_{Kr} or I_{Ks} conductance, which stabilizes membrane voltage by flattening the APD restitution curve, significantly reduced APD variability in voltage-driven instability cells (induced by increased τ_f). This occurs because APD variability increases significantly near the onset of alternans (Figs 2–4). Stabilizing Ca^{2+} cycling shifts the onset of Ca^{2+}-driven alternans but does not affect the onset of voltage-driven alternans. Therefore, Ca^{2+} stabilization more effectively suppresses APD variability in Ca^{2+}-driven instability cells than in voltage-driven instability cells, and vice versa.

This differential responsiveness of APD variability to pharmacological interventions suggests a novel approach for identifying the dominant mechanism underlying increased QTV in individual patients. By assessing the QTV response to drugs that selectively target either voltage or Ca^{2+} handling, clinicians could potentially stratify patients based on their underlying arrhythmogenic substrate. Patients exhibiting a significant QTV reduction in response to drugs stabilizing Ca^{2+} cycling might be at higher risk for Ca^{2+}-driven arrhythmias (e.g. Ca^{2+}-driven alternans, DADs, early afterdepolarizations (EADs) due to spontaneous Ca^{2+} release), while those responding to drugs stabilizing membrane voltage might be at higher risk for voltage-driven arrhythmias (e.g. voltage-driven alternans, EADs due to LTCC reactivation).

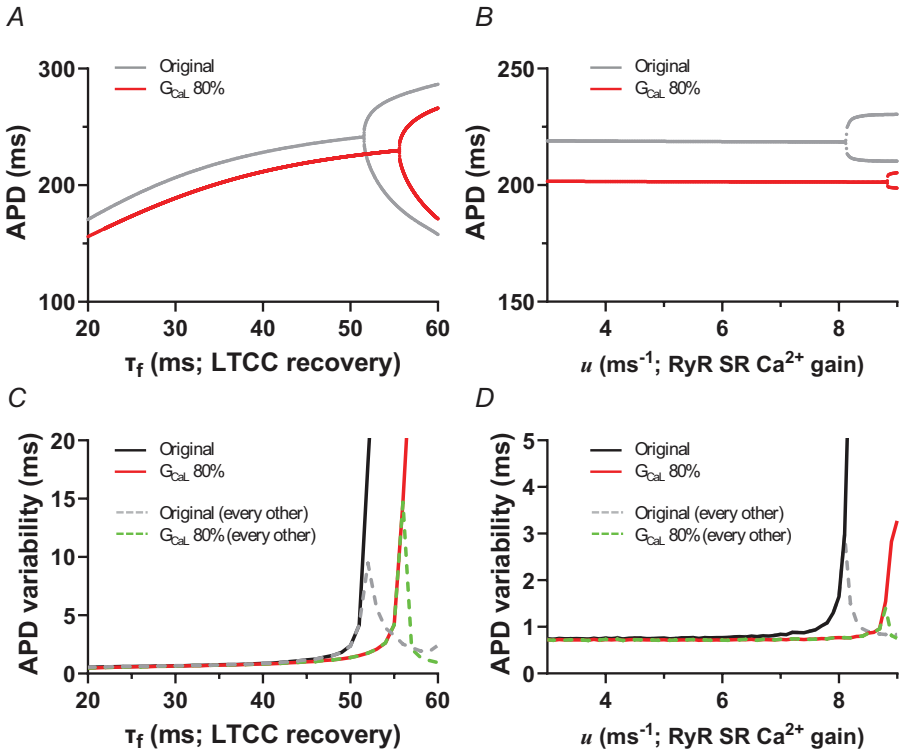

Figure 9. Bifurcation diagram and APD variability for G_{CaL} 80%
A, bifurcation diagram for voltage-driven instability. APD *vs.* τ_f. $u = 3.0$ ms^{-1}. Grey: original value; red: G_{CaL} 80%. *B*, bifurcation diagram for Ca^{2+}-driven instability. APD *vs.* u. $\tau_f = 35$ ms. *C*, APD variability *vs.* τ_f for voltage-driven instability. Black: original value; red: G_{CaL} 80%; grey: APD variability using every other beat, original value; green: APD variability using every other beat, G_{CaL} 80%. *D*, APD variability *vs.* τ_f for Ca^{2+}-driven instability. Black: original value; red: G_{CaL} 80%; grey: APD variability using every other beat, original value; green: APD variability using every other beat, G_{CaL} 80%.

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Proposed clinical strategy

The proposed clinical strategy is a pharmacological challenge test. For example, our simulation of reducing RyR conductance has direct clinical parallels. This intervention mimics the effect of RyR-stabilizing drugs such as flecainide, dantrolene or *ent*-verticilide (Do & Knollmann, 2025), which reduce spontaneous SR Ca^{2+} leak during diastole and thereby stabilize intracellular Ca^{2+} cycling. Clinically, administering one of these agents and assessing QTV is conceptually feasible. Recent experimental findings further highlight the potential for even greater diagnostic precision. As demonstrated by Uchinoumi et al., dantrolene effectively suppresses arrhythmias caused by N-terminal and central-domain RyR variants but not those in the pore domain (Uchinoumi et al., 2025). This suggests that the QTV response to dantrolene in a patient with catecholaminergic polymorphic ventricular tachycardia (CPVT) not only could indicate the presence of Ca^{2+}-driven instability but might also provide indirect information about the likely location or functional effect of the RyR mutation, adding another layer of precision to this diagnostic strategy.

In contrast, pharmacological agents that directly increase G_{Kr} and G_{Ks} are not yet in clinical use. However, this is an active area of preclinical research (Bains et al., 2024, 2025; Louradour et al., 2025; Perry et al., 2020; Schwartz et al., 2019) though these emerging strategies still face significant challenges, including selectivity, efficacy and the potential for negative inotropy, which must be overcome before clinical application. While direct pharmacological activation of I_{Kr} and I_{Ks} is not currently practical, an alternative approach is to avoid or discontinue medications that block these channels, such as dofetilide, in patients with suspected voltage-driven instability. Alternatively, if an intervention targeting Ca^{2+} cycling does not affect QTV, one could indirectly infer that the underlying mechanism is more likely voltage-driven instability.

Ultimately, the primary clinical application of our findings lies in using existing, safe drugs as diagnostic probes. While destabilizing interventions (e.g. I_{Kr} or I_{Ks} blockers, or caffeine that destabilizes RyRs) could theoretically provide similar mechanistic insights (Appendix Fig. A3), their pro-arrhythmic potential precludes their use in clinical settings. QTV analysis,

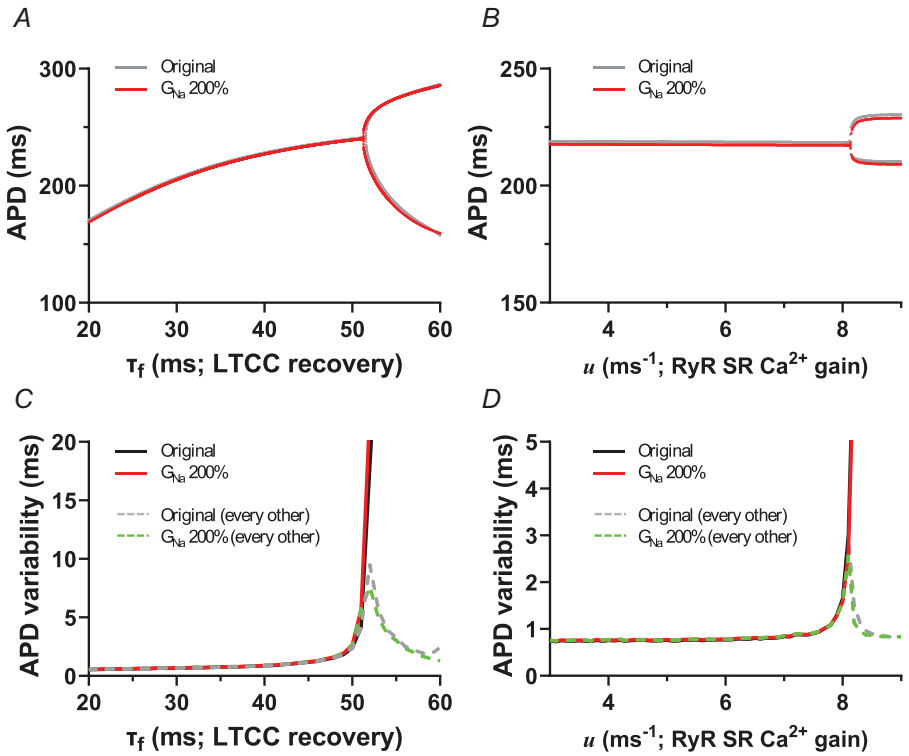

Figure 10. Bifurcation diagram and APD variability for G_{Na} 200%
A, bifurcation diagram for voltage-driven instability. APD *vs.* τ_f. $u = 3.0\ ms^{-1}$. Grey: original value; red: G_{Na} 200%. *B*, bifurcation diagram for Ca^{2+}-driven instability. APD *vs.* u. $\tau_f = 35$ ms. *C*, APD variability *vs.* τ_f for voltage-driven instability. Black: original value; red: G_{Na} 200%; grey: APD variability using every other beat, original value; green: APD variability using every other beat, G_{Na} 200%. *D*, APD variability *vs.* τ_f for Ca^{2+}-driven instability. Black: original value; red: G_{Na} 200%; grey: APD variability using every other beat, original value; green: APD variability using every other beat, G_{Na} 200%.

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combined with targeted pharmacological challenges using safe and well-tolerated drugs, offers a promising non-invasive strategy for identifying the underlying drivers of arrhythmia risk and tailoring treatment accordingly.

Limitations and future directions

While our study provides valuable insights into the relationship between APD variability and its underlying cellular mechanisms, it has several limitations.

First, our findings are based on computational modelling, and experimental validation is necessary to confirm the proposed mechanisms. Future studies should aim to validate these findings using *in vitro* and *in vivo* animal models as well as clinical studies.

Second, our model, even in 2D, does not incorporate all potential contributors to QTV, such as full 3D tissue architecture, electromechanical coupling, autonomic influences and more extensive intercellular heterogeneity. Future studies should incorporate these potential contributors to QTV. In addition, a more detailed sub-cellular representation of Ca^{2+} handling, including spatial heterogeneity of RyR clusters, could further refine our understanding of the relationship between intracellular Ca^{2+} dynamics and QTV.

Third, the pharmacological effects simulated in this study will not perfectly reflect the actions of many clinically available drugs, which can have complex pharmacokinetics and pharmacodynamics. The exact magnitude of QTV reduction in response to various pharmacological interventions remains to be determined from clinical data. Further research is required to establish clear thresholds for distinguishing between voltage-driven and Ca^{2+}-driven mechanisms based on QTV responses in patients.

Despite these limitations, our study provides a crucial mechanistic link between cellular instability and QTV. Our findings suggest that QTV can potentially serve as an early indicator of instability, even in individuals without overt arrhythmic ECG patterns. Identifying the primary drivers of QTV may enable improved risk stratification and tailored interventions for specific arrhythmia subtypes. Future studies should further investigate the clinical utility of QTV, particularly its potential to enhance arrhythmia risk assessment and management.

Conclusions

In this study, we demonstrated that instability in intra-cellular Ca^{2+} dynamics can induce Ca^{2+} transient variability independently of voltage instability, especially just before the onset of Ca^{2+} alternans. Ca^{2+} transient variability significantly contributes to APD variability via Ca^{2+}-sensitive currents. Currently, direct clinical measurement of intracellular Ca^{2+} activity remains limited. In this study, we propose that the sensitivity of QTV to specific pharmacological interventions might provide insight into the underlying mechanisms of arrhythmia risk. If QTV is significantly reduced by drugs that stabilize membrane voltage, it suggests a voltage-driven mechanism. Conversely, if QTV is significantly reduced by drugs that stabilize Ca^{2+} cycling, it suggests a Ca^{2+}-driven mechanism. Notably, these properties of QTV can be observed even in individuals without apparent arrhythmic ECG patterns, indicating that QTV may serve as an early non-invasive marker of arrhythmia risk not only in patients but also in healthy individuals.

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Appendix

Figure A1. Representative AP and Ca^{2+} transient traces.

Figure A2. 2D tissue simulations.

Figure A3. Increase of APD variability when ion channel conductances are varied.

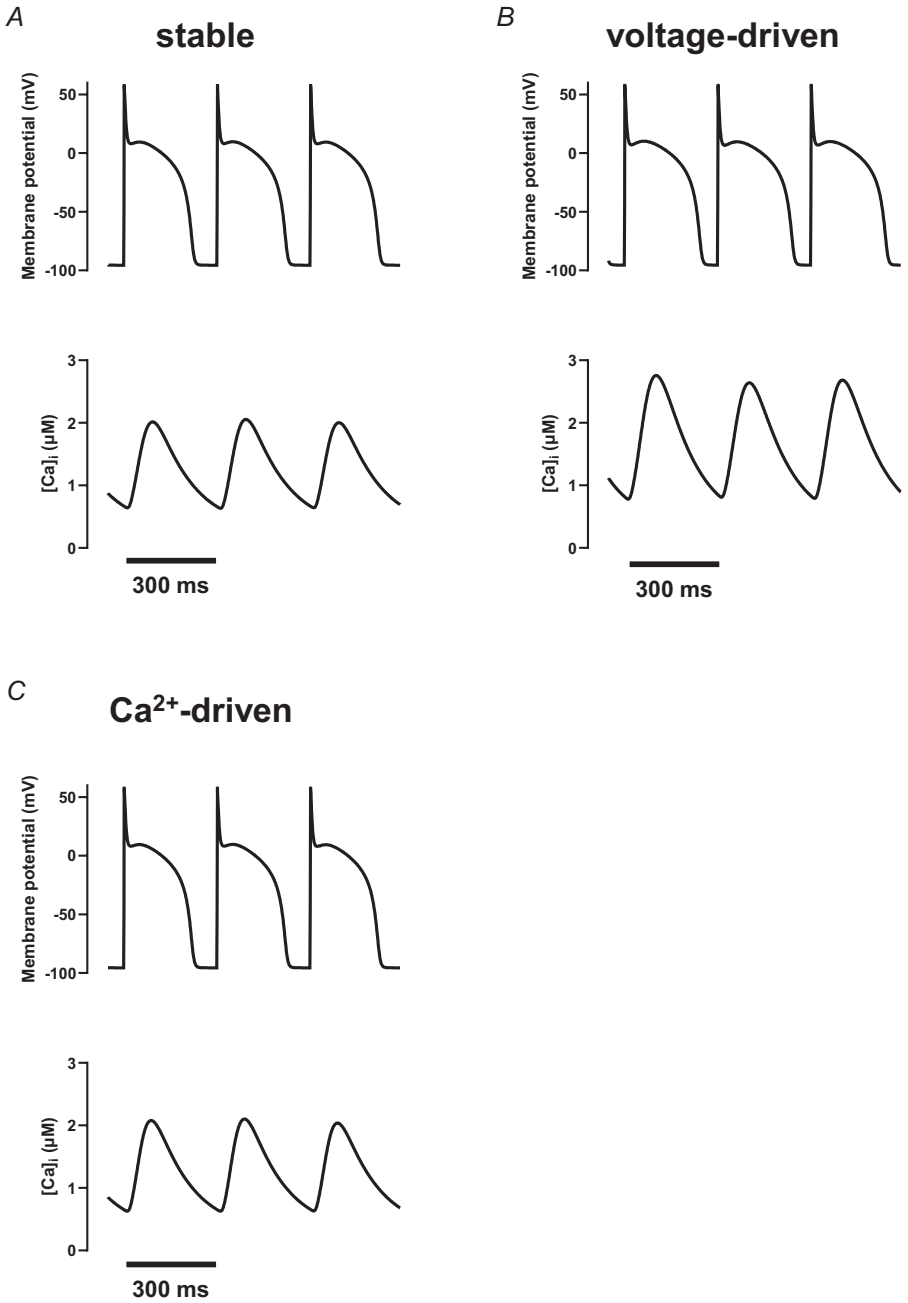

Figure A1.
A, stable condition; *B*, increased τ_f (voltage instability); *C*, increased *u* (Ca^{2+} instability).

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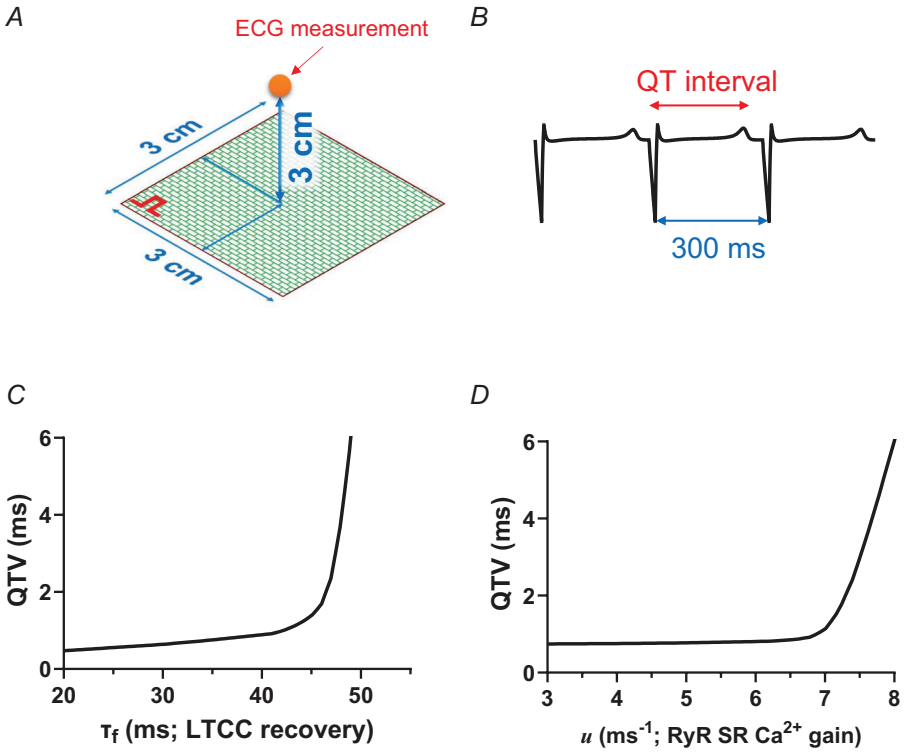

Figure A2.
A, 3 cm × 3 cm tissue was simulated. The tissue was paced from the corner. The orange circle shows the location of ECG recording. *B*, typical ECG patterns. *C*, QTV *vs.* τ_f. *D*, QTV *vs.* *u*.

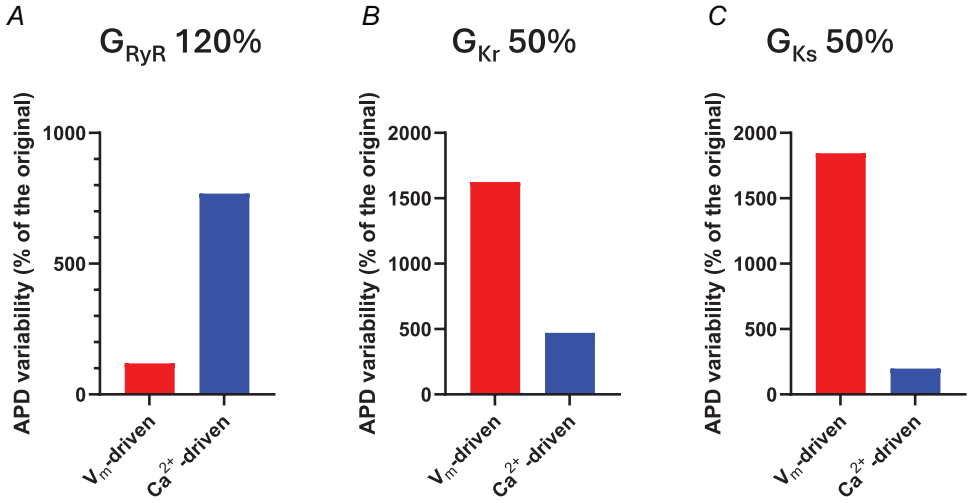

Figure A3.
A, G_{RyR} 120%. APD variability was significantly increased in Ca^{2+}-driven instability cell. *B*, G_{Kr} 50%. APD variability was significantly increased in voltage-driven instability cell. *C*, G_{Ks} 50%. APD variability was significantly increased in voltage-driven instability cell.

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Additional information

Data availability statement

The authors confirm that the data supporting the findings of this study are available within the article. The simulation codes are available on GitHub and archived on Zenodo (https://doi.org/10.5281/zenodo.16790275) and are publicly available under an MIT license.

Competing interests

The authors declare no conflicts of interest.

Author contributions

All authors contributed ideas and discussion. D.S. performed computer simulations and mathematical analysis. All authors wrote the manuscript. All authors have read and approved the final version of this manuscript and agree to be accountable for all aspects of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved. All persons designated as authors qualify for authorship, and all those who qualify for authorship are listed.

Funding

This work was supported by NIH grants (P01-HL141084, R01HL142282 to D.B.; R01HL149349 to D.S. and D.B.; R01HL111600 CMR) and American Heart Association grant 23CDA1051603 to B.H.

Keywords

alternans, intracellular calcium instability, membrane voltage instability, personalized medicine, QT interval variability

Supporting information

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